

Econometrics Project

Ala Guesmi

¹Major: BA Minor: MRK CIN: 12726292

1. Abstract

1.1. Objectives

The primary goal of this research study is to build a regression model that accurately predicts annual salaries for UAE employees, providing valuable insights for HR professionals in optimizing compensation strategies.

1.2. Methodology

The analysis follows a structured approach, outlined in the following steps:

- **Data Preprocessing:** Cleaning and preparing the dataset to ensure the quality and reliability of the data for analysis.
- **Exploratory Data Analysis (EDA):** Analyzing and visualizing the dataset to identify patterns and relationships between variables.
- **Data Transformation:** Applying transformations to improve model accuracy and prepare data for modeling.
- **Model Development:** Developing a suitable regression model by selecting relevant features and ensuring its assumptions are validated.
- **Model Evaluation:** Evaluating the model's performance using key metrics such as Adjusted R-squared, F-statistic, MSE, AIC, and BIC.
- **Model Transformation:** Applying transformations to the model, such as adding interaction terms and variables to improve its prediction accuracy.
- **Model Selection:** Selecting the best-performing model based on its adjusted R-squared and overall significance.
- **Salary Predictions:** Using the final model to predict employee salaries based on the relevant predictors.

1.3. Key Results

We obtained a final regression model that explains **92%** of the variance in **annual employee salaries**, with both **Job Role** and **Years of Experience** emerging as significant predictors.

The estimated equation:

$$Salary = 45580 + (12110 \times JobRole) + (6133.57 \times YearsExperience) \quad (1)$$

provides HR professionals with a clear, actionable formula for predicting salaries.

2. Introduction

The aim of this study is to analyze the factors that influence employee salaries in the UAE and to develop a predictive model for estimating employee annual salaries based on input features.

Understanding the determinants of salary is crucial for job seekers, companies, and policymakers, as it provides insights into compensation trends and helps in decision-making processes related to employment and salary negotiations.

3. Methodology

3.1. Dataset Description

For this analysis, we used a synthetic dataset that contains several categorical and numerical features related to employees in the UAE, including:

- **YearsExperience:** Number of years of work experience.
- **EducationLevel:** The level of education attained by the employee.
- **JobRole:** The job title or role of the employee.
- **Industry:** The industry the employee works in.
- **WorkingHoursPerWeek:** The number of hours the employee works per week.
- **Salary:** The target variable (dependent variable) representing the annual salary of employees in AED (United Arab Emirates Dirham).

Our sample dataset consists of 200 responses, with the target variable being **Salary**.

3.2. Analytical Methods and Tools Used

This analysis was conducted using Python, utilizing the following libraries and methods:

- **Pandas:** For data manipulation and preprocessing.
- **NumPy:** For numerical calculations.
- **Matplotlib/Seaborn:** For data visualization.
- **Scikit-Learn (sklearn):** For performing regression and model evaluation.
- **Statsmodels:** For statistical modeling and linear regression.
- **SciPy:** For normality testing (e.g., Jarque-Bera tests).
- **The level of significance (α)** used for hypothesis testing is set at 5%.

The primary analytical method used is **Simple Linear Regression** to predict salary based on one explanatory variable. To enhance prediction accuracy, and after a thorough evaluation, a **Multiple Linear Regression (MLR)** model was later implemented using multiple independent variables. Various model diagnostic techniques, such as assumption checking, stepwise regression, and model transformation, were employed to improve model accuracy.

4. Discussion and Results Interpretation

4.1. Data Preprocessing

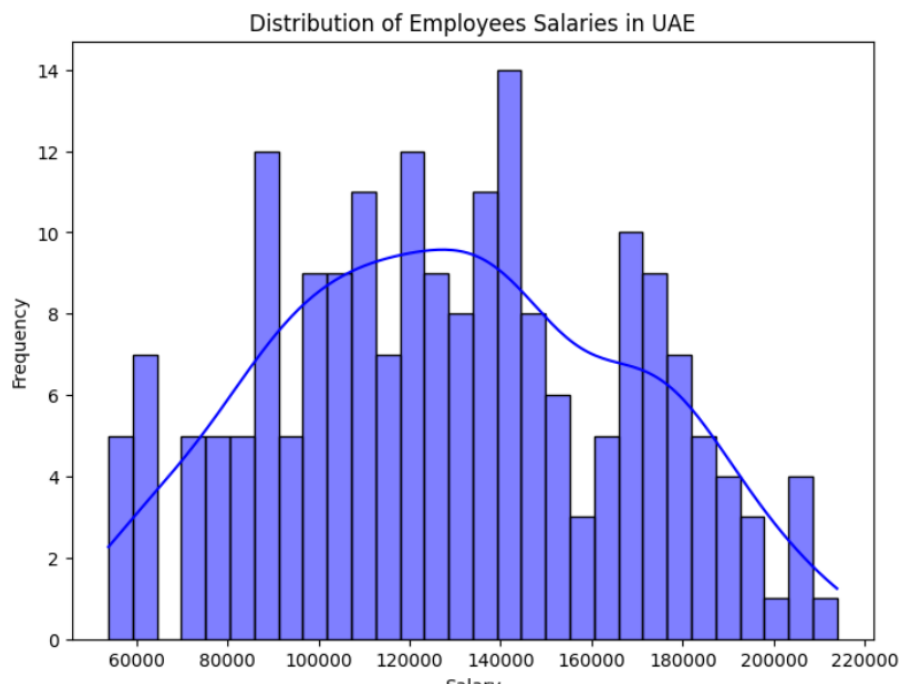
Purpose: To prepare the dataset for analysis by handling missing values, duplicates, and other inconsistencies.

- Missing values were identified and handled using the following approach:
 - Null values in categorical features were filled with the mode.
 - Null values in numerical features were filled with the mean (but no missing values were found in numerical features).
- Duplicate rows were checked and removed to ensure data quality. No duplicates were found.
- The data was cleaned to ensure it was ready for modeling.

4.2. Exploratory Data Analysis (EDA)

Purpose: To explore the structure of the dataset, summarize key characteristics, and detect relationships between features.

- The dataset's shape and size were identified:
 - 200 rows and 8 columns.
- The dataset's column types and features were inspected:
 - The number of values in each column and their specific data types (e.g., `int64`, `object`) were examined.
 - Columns were classified as categorical or numerical for further manipulation.
- A statistical summary was generated to understand the data distribution:
 - Key statistical metrics such as mean, standard deviation, quantiles, and min/max were identified for each numerical column.
- Visualizations were used to examine feature distributions and relationships between features:
 - A histogram was plotted to display the distribution of employee salaries in the UAE.



Interpretation: An amount of approximately 14,000 AED is significantly the most frequent annual salary for employees in the UAE.

4.3. Data Transformation

Purpose: To convert categorical variables into numerical representations and prepare the dataset for regression analysis.

Categorical variables such as *EducationLevel*, *JobRole*, and *Industry* were encoded into numerical format using Python’s `map()` function.

Column names were standardized for enhanced readability. Irrelevant columns were dropped, including:

- **Country:** Since the data was collected exclusively in the UAE, including this column would not provide additional insights but rather introduce redundancy.
- **ID:** As an identifier, it is irrelevant to the studied variables and the research aim.

The dataset was transformed into numerical variables suitable for modeling.

4.4. Model Building

Purpose: To build a predictive model of employees’ annual salaries in the UAE.

4.4.1. Independent Variables Identification

To select the most relevant independent variables for analyzing the dependent variable *Salary*, we focused on those with the highest correlation.

First, we generated a correlation heatmap, which is a visual representation of the relationship between numerical variables, to enhance readability and facilitate efficient selection.

We then sorted the correlation values in descending order.

Applying a correlation threshold of $|0.5|$, we identified *YearsExperience* as the most significant explanatory variable with a correlation index of approximately 0.9252.



Interpretation: The strong correlation between '*YearsExperience*' and '*Salary*' suggests that YearsExperience is a key determinant in salary predictions.

4.4.2. Fitting a Simple Linear Regression

With *Salary* as a target variable and *YearsExperience* as a predictor, we fitted a Simple Linear Regression (SLR) Model using the statsmodels library, employing the Least Squares Method (LSM).

This step is essential in providing detailed statistical information about the regression model, such as coefficients, p-values, R-squared, etc., before performing diagnostic checks.

An OLS (Ordinary Least Squares) Regression Summary was displayed:

OLS Regression Results

Dep. Variable:	Salary	R-squared:	0.856			
Model:	OLS	Adj. R-squared:	0.855			
Method:	Least Squares	F-statistic:	1177.			
Date:	Thu, 30 Jan 2025	Prob (F-statistic):	2.84e-85			
Time:	12:28:44	Log-Likelihood:	-2199.9			
No. Observations:	200	AIC:	4404.			
Df Residuals:	198	BIC:	4410.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	6.894e+04	2029.302	33.970	0.000	6.49e+04	7.29e+04
YearsExperience	6086.6375	177.389	34.312	0.000	5736.823	6436.452
Omnibus:	10.917	Durbin-Watson:	2.010			
Prob(Omnibus):	0.004	Jarque-Bera (JB):	5.758			
Skew:	-0.211	Prob(JB):	0.0562			
Kurtosis:	2.283	Cond. No.	22.7			

Interpretation:

- **Coefficients:**

- The intercept coefficient indicates that, on average, a person with no experience is expected to earn 68940 AED annually.
- The slope coefficient suggests that every additional year of experience leads to an increase in the annual salary by approximately 6086.64 AED.

- **Significance:**

- The p-value of the F-statistic confirms that the model is statistically significant.
- The low p-value for the slope coefficient further reinforces the importance of years of experience as a predictor for salary.

- **Explained Variation:**

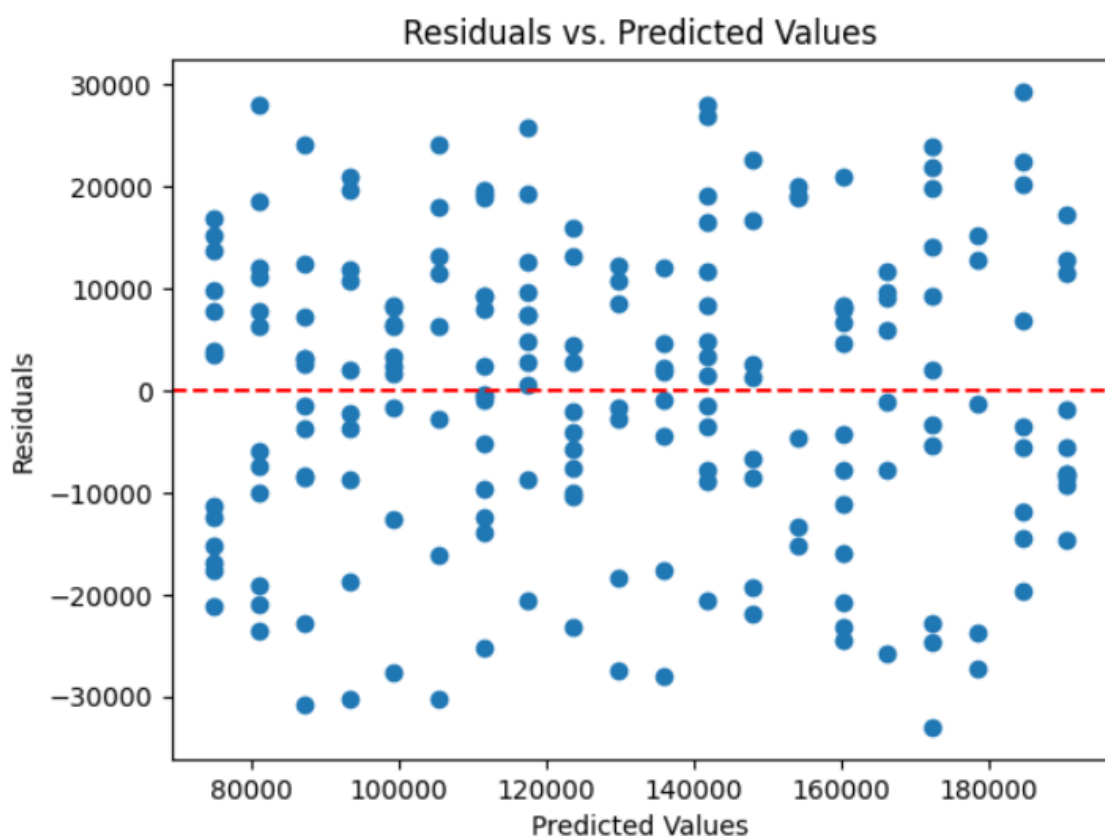
- The R-squared value of 85.6 shows that years of experience is a strong predictor of salary variation, with the model explaining a large proportion of the variation in employee salaries.

4.5. Regression Assumptions Validation

To ensure the accuracy of the model, we validated the key assumptions of regression—Linearity, Independence, Normality, and Equal variance (L.I.N.E.). To do so, we analyzed the residuals through a residual plot (Residuals vs. Fitted Values) and conducted hypothesis testing for each assumption.

Residual Analysis:

The residuals of the regression model were analyzed to check for any violations of these assumptions. A residual plot is shown below to visually inspect the relationship between the residuals and fitted values.



Interpretation: The residuals are randomly scattered around the mean (0 axis), with no visible patterns or systematic structure.

L.I.N.E Assumptions:

- **Linearity:** The relationship between the predictor and the target variable should be linear. Since the residuals do not exhibit any curved or systematic pattern, we infer that the relationship between the predictor and the dependent variable is approximately linear.
- **Independence of Errors:** The errors (residuals in our sample) should be independent of each other. The residuals do not display any clustering or trend, suggesting that errors are independent.

Since the data is not collected over time, we cannot apply the Durbin-Watson test for autocorrelation.

- **Homoscedasticity (Constant Variance of Errors):** The variance of errors (residuals) should be constant across all levels of the independent variable. The residuals appear to be evenly dispersed across all predicted values, meaning the variance remains relatively constant. If heteroscedasticity were present, we would observe a funnel shape (increasing or decreasing spread of residuals as predicted values change).

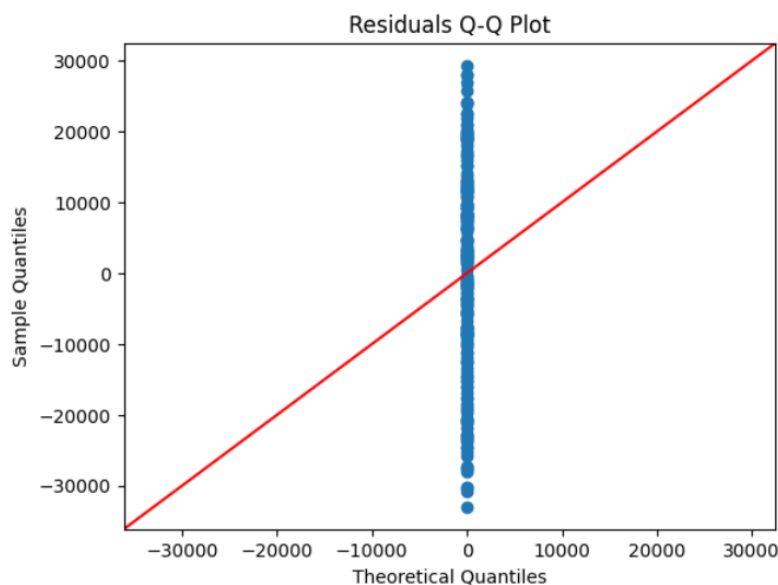
In addition to the residuals plot, we performed the **Breusch-Pagan test** for homoscedasticity:

- **Null Hypothesis (H0):** Homoscedasticity of errors.
- **Alternative Hypothesis (H1):** Heteroscedasticity of errors.

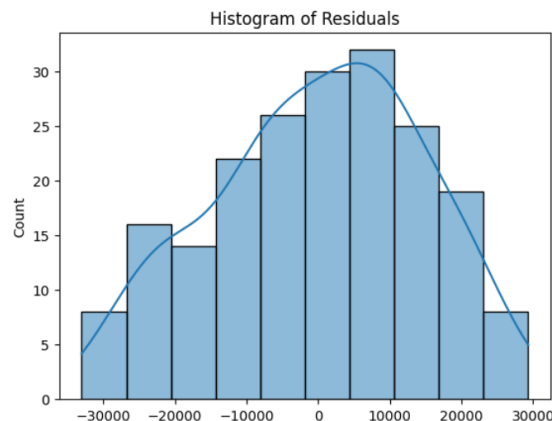
⇒ A Breusch-Pagan p-value of **0.451** indicates that we fail to reject the null hypothesis of homoscedasticity.

This means there is no significant evidence of heteroscedasticity in the model, and the residuals likely have **constant variance**.

- **Normality of Errors:** The residuals should have a normal distribution. To check residuals normality: we visualized Residuals Q-Q plot and Residuals Histogram.



Interpretation: The residuals in the Q-Q plot are stacked on top of each other along a vertical line at 0, suggesting non-normality.



Interpretation: The Residuals Histogram shows that residuals are skewed to the right and are not symmetrically distributed, which contradicts normality.

The normality assumption is checked by visualizing the residuals' distribution.

Both the Q-Q plot and the histogram indicate that the residuals follow a distribution close to normal. We also performed **Bera-Jarque test** which assumes:

- **Null Hypothesis (H0):** The data is normally distributed.
- **Alternative Hypothesis (H1):** The data is not normally distributed.

⇒ A p-value of **0.51** for the normality test indicates that we fail to reject the null hypothesis of normality.

Despite the visual indications from the Q-Q plot and histogram, the statistical test suggests that the residuals could still be approximately normal.

Possible reasons:

- The sample size may be too small for the Jarque-Bera test to detect deviations.
- The test is not always sensitive to skewness and kurtosis issues in smaller data-sets.

Therefore, we had to perform another statistical test to further check the normality assumption: the Shapiro-Wilk test which assumes:

- **Null Hypothesis (H0):** The data is normally distributed.
- **Alternative Hypothesis (H1):** The data is not normally distributed.

⇒ A p-value of **0.105** for the normality test indicates that we to reject the null hypothesis of normality.

⇒ Based on the Q-Q plot of residuals, histogram of residuals, and Shapiro-Wilk test, we can confirm that residuals do not follow a normal distribution.

LINE Assumptions Verification Conclusion:

Based on the residuals analysis, hypothesis testing, and statistical tests, we conclude that all **LINE assumptions** are **verified** except for normality. However, this violation does not necessarily compromise the model. According to the **Central Limit Theorem (CLT)**, with a sufficiently large sample size, the distribution of the estimated coefficients will

tend toward normality, meaning this deviation may not significantly impact the model's validity.

Nevertheless, we will apply transformations to the data and assess whether this issue is mitigated, ensuring the robustness of our results.

4.6. Modeling & Model Evaluation:

After verifying the assumptions of our regression model, we proceeded with the model building and evaluation process using Scikit-learn. This step focuses on training the model, assessing its performance, and ensuring it generalizes well to unseen data.

Steps Taken:

Data Splitting: We split the dataset into training (80%) and testing (20%) sets using `train_test_split` to ensure unbiased model evaluation. The training set is used to fit the model, while the testing set evaluates how well the model generalizes to new data.

Model Fitting: We trained the model using the `LinearRegression` class on the training data (`X_train`, `y_train`). This step allows the model to learn the relationship between the independent and dependent variables.

Model Evaluation: After fitting the model, we evaluated its performance on the test set (`X_test`, `y_test`). This helps assess how well the model performs on out-of-sample data, providing insight into its ability to generalize.

For evaluation, we used two metrics: **Mean Squared Error (MSE)** and **R-squared**. These metrics were compared between the train and test datasets to identify potential overfitting or underfitting issues.

Overfitting is indicated by:

- Train-MSE being significantly lower than Test-MSE. This suggests that the model performs well on the training data but struggles to generalize to unseen data (test set), typically because it has learned noise or specific patterns from the training set that do not apply to new data.

Underfitting is indicated by:

- High Train-MSE and High Test-MSE. This suggests that the model is too simple and fails to capture the underlying patterns in both the training and testing data.

Model Evaluation: MSE

Train MSE	214295628.1490957
Test MSE	191466791.33573407

⇒ Close Train-MSE and Test-MSE values suggest that the model is generalizing well, meaning it performs similarly on both the training and testing data.

Train R²	0.8531535859592128
Test R²	0.865402107780437

Model Evaluation: R²

⇒ Similar R² values for both training and testing sets indicate a well-performing model that generalizes well to new data, suggesting that the model is neither overfitting nor underfitting.

4.7. Model Transformations

Purpose: To improve the model by selecting the most relevant features and exploring possible transformations.

- **Stepwise Regression** was used to select the most important features using Mallows's Cp criterion.
- **JobRole** was found to be a significant feature but since it initially did not have a high correlation with Salary (0.2294). Therefore, YearsExperience and JobRole were combined into a Multiple Linear Regression (MLR) model.
- An **interaction term** between YearsExperience and JobRole was added to account for any joint effect these features might have on salary.
- **Interaction-Only Model:** We also tested a model that only included the interaction term to assess its unique contribution to predicting salary.

After each of the 3 new models, we displayed an **OLS Regression Summary** and we followed these two statistical metrics in comparing between models:

- **Adjusted R-squared:** Unlike regular R-squared, Adjusted R-squared accounts for the number of predictors in the model, adjusting for overfitting. It provides a more accurate measure of model performance when comparing models with different numbers of features.
- **Overall Significance test (F-Statistic):** The F-statistic tests whether the model as a whole is significant. If the overall model is not statistically significant, no conclusions can be drawn from the individual predictors, regardless of their individual significance.

⇒ The **Multiple Linear Regression Model** with '**YearsExperience**' and '**JobRole**' as predictors turned out to have the highest performance indicators compared to each of the SLR model, MLR model with an interaction term, and interaction-only model.

With enhanced adjusted R-squared of 0.92 and F-statistic of 1152 as shown in the **OLS Regression Summary** below.

OLS Regression Results

Dep. Variable:	Salary	R-squared:	0.921
Model:	OLS	Adj. R-squared:	0.920
Method:	Least Squares	F-statistic:	1152.
Date:	Thu, 30 Jan 2025	Prob (F-statistic):	1.94e-109
Time:	15:31:19	Log-Likelihood:	-2139.6
No. Observations:	200	AIC:	4285.
Df Residuals:	197	BIC:	4295.
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	4.558e+04	2368.200	19.249	0.000	4.09e+04	5.03e+04
JobRole	1.211e+04	948.360	12.770	0.000	1.02e+04	1.4e+04
YearsExperience	6133.5679	131.593	46.610	0.000	5874.057	6393.079

Omnibus:	9.336	Durbin-Watson:	2.144
Prob(Omnibus):	0.009	Jarque-Bera (JB):	5.552
Skew:	-0.234	Prob(JB):	0.0623
Kurtosis:	2.331	Cond. No.	37.8

Model Fit:

- After adjusting for the number of predictors, the variation in **JobRole** and **YearsExperience** explains **92%** of the variation in employee annual salary in the UAE.
- The p-value of the F-statistic is very low (**1.9e-109**), indicating a high overall significance of the model.

Coefficients:

- **Intercept:** 45,580 AED
- **JobRole:** 12,110 AED – With an upgrade in the employee job role, the average annual salary is estimated to increase by **12,110 AED**, holding the **YearsExperience** variable constant.
- **YearsExperience:** 6,133.57 AED – With one additional year of experience, the average annual salary is estimated to increase by **6,133.57 AED**, holding the **JobRole** variable constant.

Significance:

- All coefficients have very low p-values (< **0.05**), indicating that they are statistically significant.
- The confidence intervals of all predictors do not include zero, further confirming their significant effect on **Salary**.

Multicollinearity:

- After calculating the **Variance Inflation Factor (VIF)**, both predictors showed a VIF index close to 1, indicating **no multicollinearity** between independent variables.

Model Selection Metrics:

- **AIC:** 2490, **BIC:** 2506 – Lower values of these metrics indicate a better model fit.

Estimated Model Equation:

$$\text{Salary} = 45580 + (12110 \times \text{JobRole}) + (6133.57 \times \text{YearsExperience})$$

4.8. Prediction

Below is a representation of the fitted values.



-We tried our predictive model on a set of values after making sure that we do not exceed the range of both of our predictors 'JobRole' and 'YearsExperience': [1,3] and [1,20], respectively.

-The model has successfully provided us with predictive values of annual employee salary as shown below:

-The model has successfully provided us with predictive values of annual employee salary as shown below:

	JobRole	YearsExperience	Predicted Salary
0	3	12	155519.050208
1	1	2	69962.082966
2	2	8	118874.134508

5. Key Considerations

We wanted to further recheck the normality assumption with our final model (MLR); however, the residuals histogram still demonstrates right skewness, indicating that non-normality persists. To address this issue and suggest possible solutions, we considered the following:

Sample Size

Increasing the sample size can help the residuals' distribution approach normality, according to the Central Limit Theorem.

Robust Regression

An alternative approach is to use Robust Regression, which is less sensitive to violations of assumptions like normality and heteroscedasticity. This method can be applied if the normality issue persists and significantly impacts the model.

⇒ While the normality violation is acknowledged, it does not critically affect overall model performance, especially given the sample size and the robustness of the regression techniques used.

6. Actionable Insights

Key Drivers of Employee Salary

Job Role and Years of Experience are the key factors explaining 92% of the variation in employee salary. This suggests that these two variables are strong predictors, making them highly valuable for HR or compensation strategy decisions.

⇒ When designing compensation structures, prioritize job roles and years of experience as these directly impact salary levels.

Impact of Job Role

A one-step upgrade in a job role leads to an estimated 12,110 AED increase in salary, assuming years of experience is constant.

⇒ Consider setting clear salary increments for employees who advance in job roles to ensure alignment with the model's prediction. This will help retain employees and motivate role progression.

Impact of Experience

For each additional year of experience, employees' salaries are estimated to increase by 6,133.57 AED.

⇒ Introduce a structured pay scale that rewards employee experience. This will ensure salary growth is aligned with experience and can guide promotion and retention strategies.

7. Conclusion

This study successfully analyzed the factors influencing the **annual salary** of employees in the UAE, using a cleaned dataset of 200 employees and focusing on **Job Role** and **Years of Experience** as key predictors. The model showed that these two variables explain **92%** of salary variation, with both demonstrating strong statistical significance. Evaluation metrics, including **Adjusted R-squared**, **F-statistic**, and **AIC/BIC**, confirmed the model's robustness and good fit. Data visualization further supported the interpretation of these relationships. Despite some violations of the normality assumption in the residuals, the model performed well in predicting salary levels.

Future improvements could involve using robust regression techniques or increasing the sample size to address normality violations.

Overall, this analysis provides valuable insights into the factors that influence salaries in the UAE and offers a reliable tool for salary prediction and HR professionals in the UAE.

8. Recommendations

- **Salary Benchmarking:** Use this model for benchmarking employee salaries against industry standards and ensure competitiveness in attracting and retaining talent.
- **Regular Model Review:** Periodically review the model's accuracy with new data to ensure it stays relevant over time, especially as industry trends and salary expectations change.
- **Study Expansion:** Expand this research study to cover other countries and obtain a more robust predictive model customized for each country, enabling organizations to design more accurate, data-driven salary strategies tailored to diverse markets.

article hyperref

9. References

- **Datasets:** <https://www.kaggle.com/code/rtatman/datasets-for-regression-analysis/notebook>
- **Employee Salary UAE Dataset:** <https://drive.google.com/uc?id=1csiDOrRwE6ONbF6Iov349eaGr>